Study BM or Gaussian processes before!

Existence of Brownian Motion

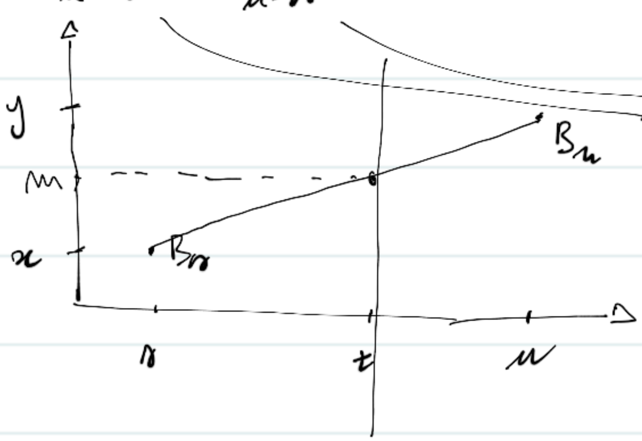
Lévy's construction.

There is a useful result we should prove in advance, which helps the intuition.

Proposition. Let $(B_t)_{t\geq 0}$ be a 1-dim BM and $0\leq s<t<u$. Then

$$A \mapsto P(B_t \in A \mid B_s = x,\ B_u = y) = N(m, \sigma^2)$$

$$m := \frac{u-t}{u-s}x + \frac{t-s}{u-s}y \qquad \sigma^2 := \frac{(u-t)(t-s)}{u-s}$$



"weights" for x and y: if $u-t = t-s$ then $m = \frac{x+y}{2}$

m is on the line!!

$$B_t \sim N(m, \sigma^2)$$

Proof. Use that the vector (B_s, B_t, B_u) is Gaussian with mean $\underline{0}$ and covariance matrix C given by

$$C := \begin{bmatrix} s & s & s \\ s & t & t \\ s & t & u \end{bmatrix}$$

(Prove this as an exercise).

Use that if $(X\,Y\,Z)$ is a random vector with density $p_{XYZ}(x\,y\,z)$ and $p_{YZ}(y,z)$ is the marginal of (Y,Z), then

$$P(X \in B \mid Y=y,\ Z=z) = \int \mathbb{1}_B(x)\, p_{X|Y,Z}(x)\, dx$$

where

$$p_{X|Y,Z}(x) = \frac{p_{XYZ}(x,y,z)}{p_{YZ}(y,z)}.$$

In our case the distribution of $(B_s\,B_t\,B_u)$ is

$$p_{s\,t\,u}(x\,y\,z) = \frac{1}{(2\pi)^{3/2}} \frac{1}{\sqrt{(u-t)(t-s)\,s}} \exp\left(-\frac{1}{2}\left[\frac{(z-y)^2}{u-t} + \frac{(y-x)^2}{t-s} + \frac{x^2}{s}\right]\right)$$

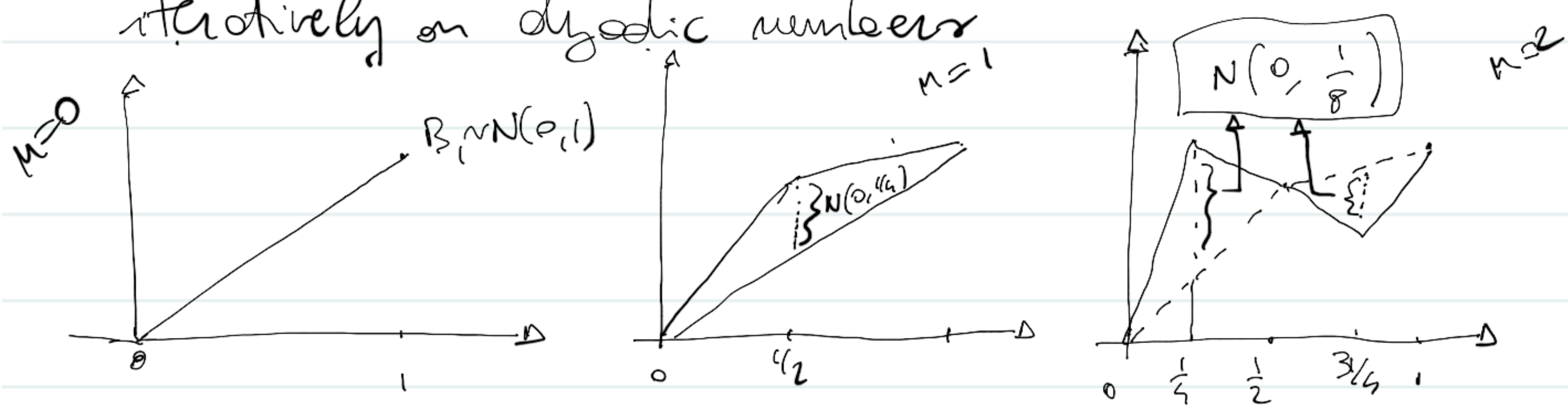
and

$$p_{s,u}(x,z) = \frac{1}{(2\pi)} \frac{1}{\sqrt{(u-s)\,s}} \exp\left(-\frac{1}{2}\left[\frac{(z-x)^2}{u-s} + \frac{x^2}{s}\right]\right).$$

Do this as an exercise.

$\lrcorner$

The idea of the Lévy construction is to define BM iteratively on dyadic numbers



By the previous proposition we have that

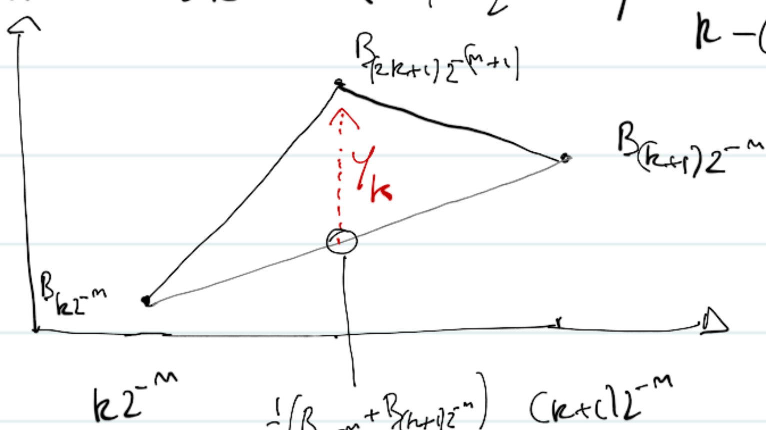
$$A \mapsto P\left(B_{(2k+1)2^{-(m+1)}} \in A \,\middle|\, B_{k2^{-m}} = x,\ B_{(k+1)2^{-m}} = y\right) = N\left(\tfrac{1}{2}(x+y), \tfrac{1}{4} 2^{-m}\right)$$

Here is the Algorithm:

1°. Set $B_0 = 0$ and $B_1 \sim N(0,1)$

2°: Assume we have already defined $B(k2^{-m})$, $k = 0 \dots 2^m$, such that $B_{(k+1)2^{-m}} - B_{k2^{-m}} \left(\sim N(0, 2^{-m})\right)$ and are independent $(k = 0 \dots 2^m - 1)$

3°: Denote Y_k the distance between the interpolation of $B_{(k+1)2^{-m}}$ and $B_{k2^{-m}}$ at the point $(2k+1)2^{-(m+1)}$ and suppose $Y_k \sim N\left(0, \frac{1}{2^{m+2}}\right)$ and are independent $k = (0 \dots 2^m - 1)$



$Y_k \sim N\left(0, \frac{1}{2^{m+2}}\right)$

$k = 0 \dots 2^m - 1$

and define now

$$B_{e2^{-(m+1)}} := \begin{cases} B_{k2^{-m}} & e = 2k \quad (\text{even}) \\ \frac{1}{2}\left(B_{k2^{-m}} + B_{(k+1)2^{-m}}\right) + Y_k & \boxed{e = 2k+1 \quad (\text{odd})} \end{cases}$$

→ new dyadic when m increases

Remark. Take the increments $B_{(2k+1)2^{-(m+1)}} - B_{(2k)2^{-(m+1)}} =$

$$= \frac{1}{2}\left(B_{k2^{-m}} + B_{(k+1)2^{-m}}\right) + Y_k - B_{k2^{-m}} = \underbrace{\frac{1}{2}\left(B_{(k+1)2^{-m}} - B_{k2^{-m}}\right) + Y_k}$$

and note that they are again Gaussian and indep. by

$\mathrm{Var}(\ ) = \frac{1}{4} \cdot \frac{1}{2^m} + \frac{1}{2^{m+2}} = \frac{1}{2^{m+1}}$

properties of Gaussian r.v.'s:

If $(X_k)_{k=1\dots N}$ $(Y_k)_{k=1\dots N}$ are such that

- $X_k \sim N(0,\sigma^2)$ $Y_k \sim N(0,\sigma^2)$
- $X_1 \dots X_N$ are indep $Y_1 \dots Y_N$ are indep.
- $(X_k)_{k=1\dots N}$ and $(Y_k)_{k=1\dots N}$ are indep.

Then $X_1+Y_1, X_1-Y_1, \dots, X_N+Y_N, X_N-Y_N$ are indep and $X_k \pm Y_k \sim N(0, 2\sigma^2)$ $k=1\dots N$.

To apply this in our situation take $X_k = \frac{1}{2}(B_{(2k+1)2^{-m-1}} - B_{k2^{-m}})$
$Y_k = Y_k$

Here is the result by Lévy.

Theorem (Lévy 1940) For $m \geq 0$, let $W_m(t,\omega)$ be the linear interpolation of $B_{k2^{-m}}$ $k=0\dots 2^m$. Then the series

$$\bar{B}(t,\omega) := \sum_{m=0}^{\infty} \left(W_{m+1}(t,\omega) - W_m(t,\omega)\right) + W_0(t,\omega) \qquad t\in[0,1]$$

converges a.s. uniformly in $t\in[0,1]$. The process $(\bar{B}_t)_{t\in[0,1]}$ is (indistinguishable from) a BM

Note that

$$\bar{B}(t,\omega) := \sum_{m=0}^{\infty} \left(W_{m+1}(t,\omega) - W_m(t,\omega)\right) + W_0(t,\omega)$$

$$= \lim_{N\to\infty} \sum_{m=0}^{N} \left(W_{m+1}(t,\omega) - W_m(t,\omega)\right) + W_0(t,\omega)$$

$$= \lim_{N\to\infty} W_N(t,\omega)$$

Definition. Let $(X_t)_{t\ge 0}$, $(Y_t)_{t\ge 0}$ be real valued random processes on a given $(\Omega, \mathcal{A}, P)$.

1) The processes are called indistinguishable if there is a null set $\Omega_0 \subset \Omega$ s.t.

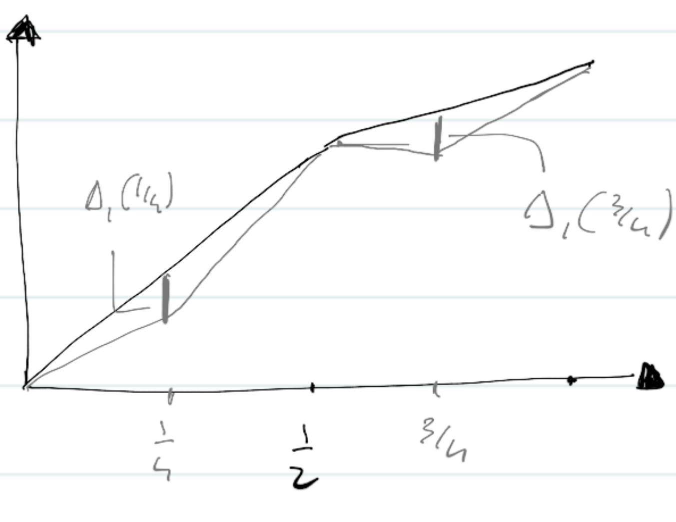
$$X_t(\omega) = Y_t(\omega) \quad \forall \omega \notin \Omega_0,\ \forall t \ge 0$$

2) The processes are called versions of each other if $\forall t \ge 0$ there exist a null set $\Omega_0(t) \subset \Omega$ s.t.

$$X_t(\omega) = Y_t(\omega) \quad \forall \omega \notin \Omega_0(t)$$

Proof. Set $\Delta_m(t,\omega) := W_{m+1}(t,\omega) - W_m(t,\omega)$.
By construction, the maximal distance is attained at one of the midpoints of the intervals $[k2^{-m}, (k+1)2^{-m}]$:

$$\sup_{t\in[0,1]} |\Delta_m(t,\omega)| \le \sup_{1\le k\le 2^m} |\Delta_m((2k-1)2^{-m-1}, \omega)|$$



$m = 2$

We show now that

$$\sum_{m=1}^{\infty} \sup_{1\le k\le 2^m} \left| \Delta_m\left((2k-1)2^{-m-1}, \omega\right)\right| < \infty$$

on a set $\Omega_0 \subset \Omega$ with $P(\Omega_0) = 1$. This implies (a.s.) uniform convergence of the series.

Let y_m be such that $\sum_m y_m < \infty$.

Note that

$$\sum_m P\Big(\sup_{1\le k\le 2^m} \big|\underbrace{\Delta_m\left((2k-1)2^{-m-1}\right)}_{\sim N(0,\, 2^{-(m+2)})}\big| > y_m\Big)$$

these are the ξ_{k}

$$\le \sum_m P\Big(\big|\underbrace{\sqrt{2^{m+2}}\,\Delta_m\left((2k-1)2^{-m-1}\right)}_{N(0,1)}\big| > \sqrt{2^{m+2}}\, y_m\Big)$$

$$P(\sup >) = P\Big(\bigcup_{k=1}^{2^m} >\Big) \le \sum_{k=1}^{2^m} P(\,) =$$

$$\le 2^m \, 2 \int_{\sqrt{2^{m+2}}\, y_m}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\, dy.$$

$$\le 2^{m+1} \int_{\sqrt{2^{m+2}}\, y_m}^{\infty} \underbrace{\frac{y}{\sqrt{2^{m+2}}\, y_m}}_{>1} \frac{e^{-y^2/2}}{\sqrt{2\pi}}\, dy$$

$$= 2^{m+1} \frac{1}{\sqrt{2^{m+2}}\, y_m} \frac{1}{\sqrt{2\pi}} e^{-\frac{y_m^2 2^{m+2}}{2}}$$

choose $y_m := \dfrac{c\sqrt{2m \log 2}}{\sqrt{2^{m+2}}}$ for some $c > 1$

$$= \frac{2^{m+1}}{\sqrt{2\pi}} \frac{1}{c\sqrt{2m\log 2}} e^{-c^2 2m\log 2}$$

$$= \frac{1}{\sqrt{\pi}} \frac{1}{c} \frac{1}{\sqrt{m}} \frac{1}{\sqrt{\log 2}} \frac{2^m}{2^{2mc^2}} \leq \frac{1}{\sqrt{m\log 2}} \frac{1}{2^{m(2c^2-1)}}$$

and so

$$\sum_m P\Big(\underbrace{\sup_{1\leq k\leq 2^m} \left|\Delta_m((2k-1)2^{-m-1})\right| > y_m}_{=: A_m}\Big) < \infty.$$

By the 1^{st} B-C lemma $P(\limsup A_m) = 0 \to P(\underbrace{\liminf A_m^c}_{=: \Omega_0}) = 1$
hence there exist $\Omega_0 \subset \Omega$ with $P(\Omega_0) = 1$ s.t.

$\forall \omega \in \Omega_0 \ \exists N(\omega) \in \mathbb{N} : \forall m > N(\omega)$ it is true that

$$\sup_{1\leq k\leq 2^m} \left|\Delta_m((2k-1)2^{-m-1})\right| \leq y_m = \frac{c\sqrt{2m\log 2}}{\sqrt{2^{m+2}}}$$

It follows that $\lim_{N\to\infty} W_N(t,\omega)$ exists for all $\omega\in\Omega_0$ $t\in[0,1]$, and the series converges uniformly in $t\in[0,1]$.
In particular, the paths $t\mapsto B(t,\omega)$ are continuous for all $\omega\in\Omega_0$ by uniform continuity.
Now we prove that

$$\tilde{B}(t,\omega) := \begin{cases} \bar{B}(t,\omega), & \omega\in\Omega_0, \\ 0, & \omega\notin\Omega_0, \end{cases} \quad t\in[0,1],$$

is a Brownian motion.
It is clear that $\tilde{B}(0) = 0$ and that $(\tilde{B}_t)_{t\in[0,1]}$

has continuous sample paths.

Stationarity:

Let $0 \le s \le t$ be two dyadic numbers, i.e., $s = j 2^{-m}$ $t = k 2^{-m}$ for some $j < k$ and $m \in \mathbb{N}$.

$$\tilde{B}(k 2^{-m}) - \tilde{B}(j 2^{-m}) \overset{a.s.}{=} \bar{B}(k 2^{-m}) - \bar{B}(j 2^{-m})$$

$$= W_m(k 2^{-m}) - W_m(j 2^{-m})$$

$$= \underbrace{\sum_{\ell=j+1}^{k} \underbrace{\left(B(\ell 2^{-m}) - B((\ell-1) 2^{-m})\right)}_{\text{indep for } \ell = j+1, \dots, k}}_{\sim N(0, t-s)}$$

$$\longrightarrow \underbrace{B(\ell 2^{-m}) - B((\ell-1) 2^{-m})}_{N(0, 2^{-m})}$$

$$\sum_{\ell=j+1}^{k} 2^{-m} = \frac{k-j}{2^m}$$

For general $s, t \in [0,1]$ take two sequences of dyadic numbers s_k t_k with $t_k \downarrow t$, $s_k \downarrow s$. Then

$$E e^{i\xi(\tilde{B}(t) - \tilde{B}(s))} = \lim_{k\to\infty} E \exp\left(i\xi(\tilde{B}(t_k) - \tilde{B}(s_k))\right) \longrightarrow \text{Bounded Convergence}$$

$$= \lim_{k\to\infty} e^{-\frac{t_k - s_k}{2}\xi^2} = e^{-\frac{t-s}{2}\xi^2}$$

Independence follows analogously, using indep. of increments on the dyadic numbers.

$$E e^{i\xi[\tilde{B}(t^1) - \tilde{B}(t^0)][\tilde{B}(s^1) - \tilde{B}(s^0)]} = \lim_{k}$$

$$= \dots$$

$$= E e^{i\xi[\tilde{B}(t^1) - \tilde{B}(t^0)]} \, E e^{i\xi[\tilde{B}(t^1) - B(t^0)]}$$

Now we use this result to construct a 1-dim BM on $t \geq 0$.

Corollary. There is a 1-dim BM $(B_t)_{t\geq 0}$.

Proof. Let $(\Omega_k, \mathcal{A}_k, P_k)$, $k \in \mathbb{N}_0$, be probability spaces. Use the previous Theorem to construct BMs $(B^k_t)_{t\in[0,1]}$ on $(\Omega_k, \mathcal{A}_k, P_k)$. Consider the product space
$$(\Omega, \mathcal{A}, P) := \bigotimes_{k\in\mathbb{N}_0} (\Omega_k, \mathcal{A}_k, P_k).$$
Define, on Ω, the stochastic processes
$$W^k(t,\omega) := B^k(t, \omega_k), \qquad t \in [0,1],\ k \in \mathbb{N}_0$$
where $\Omega \ni \omega = (\omega_0, \omega_1, \dots, \omega_k, \dots)$

The processes $(W^k(t))_{t\in[0,1]}$, $k \in \mathbb{N}_0$, are independent BMs on Ω.

Now define
$$B_t := \begin{cases} W^0(t) & t \in [0,1) \\ W^0(1) + W^1(t-1) & t \in [1,2) \\ \sum_{k=0}^{m-1} W^k(1) + W^m(t-m) & t \in [m, m+1),\ m \in \mathbb{N}_0 \end{cases}$$

We prove that $(B_t)_{t\geq 0}$ is a BM on $(\Omega, \mathcal{A}, P)$.

Continuity of paths is clear.

Stationarity of increments: let $s < t$ be such that $s \in [\ell, \ell+1)$ and $t \in [m, m+1)$ for some $\ell \leq m$.

If $\ell = m$, by definition
$$B_t - B_s = W^m(t-m) - W^m(s-m) \sim N(0, t-s)$$

If $\ell < m$, we have

$$B_t - B_s = \left(\sum_{k=0}^{m-1} W^k(1) + W^m(t-m)\right) - \left(\sum_{k=0}^{\ell-1} W^k(1) + W^\ell(s-\ell)\right)$$

$$= \sum_{k=\ell}^{m-1} W^k(1) + W^m(t-m) - W^\ell(s-\ell)$$

$$= \underbrace{\underbrace{W^\ell(1) - W^\ell(s-\ell)}_{\substack{\sim N(0,\, 1-(s-\ell)) \\ \text{since } W^\ell \text{ is BM}}} + \underbrace{\sum_{k=\ell+1}^{m-1} W^k(1)}_{\substack{\sim N(0,\, m-\ell-1) \\ \text{since } W^k \text{ are indep } N(0,1)}} + \underbrace{W^m(t-m)}_{\sim N(t-m)}}_{\sim N(1-(s-\ell)+m-\ell-1+t-m)}$$

$$\sim N(1-(s-\ell)+m-\ell-1+t-m) = t-s$$

We check independence of increments (let's do this only for two increments. Take $s < t < u$ with $s \in [\ell, \ell+1)$, $t \in [m, m+1)$ and $u \in [n, n+1)$. Then

$$B_u - B_t = \begin{cases} W^m(u-m) - W^m(t-m) & m = n \\ W^m(1) - W^m(t-m) + \sum_{k=m+1}^{n-1} W^k(1) + W^n(u-n) & m < n \end{cases}$$

We wrote before the increment $B_t - B_s$. The r.v. appearing in the representations for $B_u - B_t$ and $B_t - B_s$ are indep. and thus $B_t - B_s$ and $B_u - B_t$ are independent. $\blacksquare$

The canonical Brownian Motion

So far we constructed a process $(B_t)_{t\geq 0}$ which is a BM on a given probability space. The canonical construction of BM, instead, aims to construct a probability space such that a given process (the canonical process) is a BM.

Let $I := [0,\infty)$ and

$$C_{(0)}(I) := \{ w : I \to \mathbb{R} : w \text{ is continuous and } w(0) = 0 \}.$$

The canonical process, $(X_t)_{t\geq 0}$, is defined by

$$X_t(w) := w(t), \qquad w \in C_0(I).$$

So our goal is to find a probability measure μ on (a suitable σ-Algebra on) $\Omega = C_{(0)}(I)$ s.t. $(X_t)_{t\geq 0}$ is a BM. Which σ-Algebra do we choose?

→ Consider $C_{(0)}(I)$ as a metric space with the metric $\rho(v,w) := \sum_{n\geq 1} 2^{-n} \min\{ \sup_{0\leq t\leq n} |v(t)-w(t)|, 1 \}$.

One can prove that, by definition,

$$\rho(v_n, v) \to 0 \iff v_n \to v \text{ uniformly on compact sets}$$

$$\left(\text{i.e., } \forall T>0,\ \lim_{n\to\infty} \sup_{t\in[0,T]} |v_n(t) - v(t)| = 0\right)$$

It is also true that the metric space is separable, i.e., there exists a countable dense subset. The Borel σ-Algebra on this space is the smallest σ-Algebra containing the open sets $U \subset C_{[0]}$, i.e., the sets U s.t. for every $u \in U$ there exists $r > 0$ s.t.

$$\underbrace{B_\rho(u,r) := \{ v \in C_0(I) \, ; \, \rho(u,r) < r \}}_{\text{open ball in the metric space } (C_{[0]}(I), \rho)} \subset U.$$

→ Another possibility is to take, on the space

$$\mathbb{R}^I = \{ w : I \mapsto \mathbb{R} \}$$

the product σ-Algebra, i.e., → $\bigotimes_{t \in I} \mathcal{B}(\mathbb{R}) := \sigma(\text{measurable } \underbrace{\text{rectangle}}_{\{w \in \mathbb{R}^I : w(t) \in A_t\},\ A_t \neq \mathbb{R} \text{ for a finite number of } t,\ A_t \in \mathcal{B}(\mathbb{R})})$

$$\mathcal{B}^I(\mathbb{R}) := \sigma\left(\pi_t^{-1}(B) : B \in \mathcal{B}(\mathbb{R}),\ t \geq 0 \right)$$
$$= \sigma(\pi_t : t \geq 0)$$

generated by the projections $\pi_t : \mathbb{R}^I \mapsto \mathbb{R}$ $w \mapsto w(t)$.
Since $C_0(I) \subset \mathbb{R}^I$ the trace of $\mathcal{B}^I(\mathbb{R})$ on $C_0(I)$ is

$$C_{[0]}(I) \cap \mathcal{B}^I(\mathbb{R}) := \{ B \cap C_{[0]}(I) \, ; \, B \in \mathcal{B}^I(\mathbb{R}) \}$$

Recall: the trace is a σ-Algebra.
What is the best choice?

Lemma. $\mathcal{B}(C_{[0]}(I)) = C_{[0]}(I) \cap \mathcal{B}^I(\mathbb{R})$.

Proof. The mapping $C_{[0]} \ni w \mapsto \pi_t(w) = w(t)$

is continuous wrt ρ. Indeed if $t \in [0, M]$ for some M, then

$$\min\{1, |\pi_t(w) - \pi_t(v)|\} = \min\{1, |w(t) - v(t)|\}$$
$$\leq \min\{1, \sup_{s \leq M} |w(s) - v(s)|\}$$
$$= 2^M 2^{-M} \min\{1, \sup_{s \leq M} |w(s) - v(s)|\}$$
$$\leq 2^M \sum_k{}' 2^{-k} \min\{1, \sup_{s \leq k} |w(s) - v(s)|\}$$
$$= 2^M \rho(v, w)$$

It follows that, $\forall \varepsilon \; \exists \delta_\varepsilon$:

$$\rho(v, w) < \delta_\varepsilon \longrightarrow |\pi_t(v) - \pi_t(w)| < \varepsilon.$$

Since ε ... $\Rightarrow |\pi_t(w) - \pi_t(v)| \leq 2^M \rho(v, w) \leq 2^M \frac{\varepsilon}{2^M} \Rightarrow \delta_\varepsilon = \frac{\varepsilon}{2^M}$

Therefore $\pi_t : (C_0(I), \mathcal{B}(C_0(I))) \longmapsto (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable since any continuous function between (Borel) measure space is measurable. It follows that

$$C_0(I) \cap \mathcal{B}^I(\mathbb{R}) = \sigma(\pi_t|_{C_0(I)} : t \geq 0) \subset \mathcal{B}(C_0(I)).$$

Now we prove the opposite inclusion.

Since $(C_0(I), \rho)$ is a separable metric space then the Borel σ-Algebra is generated by the open balls $B_\rho(w, r)$, $w \in C_0(I)$, $r > 0$. Therefore it is sufficient to show that $B_\rho(w, r) \in C_0(I) \cap \mathcal{B}^I(\mathbb{R})$ for all $w \in C_{(0)}$ and $r > 0$. We have

$$\rho(v, w) = \sum_{M \geq 1}{}' 2^{-M} \min\{1, \sup_{0 \leq t \leq M} |\pi_t(w) - \pi_t(v)|\}$$

$$= \sum_{M \geq 1}{}' 2^{-M} \min\{1, \sup_{t \in [0, M] \cap \mathbb{Q}} |\pi_t(w) - \pi_t(v)|\}$$

so that

(by properties of measurable functions)

$$C_0(I) \ni v \mapsto F(v) := \rho(v, w) \in \mathbb{R}$$ is measurable

w.r.t. $\sigma(\tilde{\pi}_t|_{C_0(I)} : t \geq 0) = \mathcal{B}^I(\mathbb{R}) \cap C_{C_0}(I)$.

RECALL $\tilde{\pi}_t$ IS MEAS!

Therefore

$$B_\rho(w, r) = F^{-1}([0, r)) \in \mathcal{B}^I(\mathbb{R}) \cap C_0(I)$$

We remark that, by definition, the sets

$$\{\tilde{\pi}_{t_1} \in A_1, \dots, \tilde{\pi}_{t_n} \in A_n\}, \quad t_i \in I, A_i \in \mathcal{B}(\mathbb{R}), n \in \mathbb{N}$$

are a generator of $\mathcal{B}^I(\mathbb{R})$. This implies the following:

Corollary. Every probability measure on $(C_0(I), \mathcal{B}(C_{C_0}(I)))$ is uniquely determined by its values on the cylinder sets

$$\{w \in C_0(I) : w(t_1) \in A_1, \dots, w(t_n) \in A_n\}, \quad t_i \in I, \; A_i \in \mathcal{B}(\mathbb{R}), \; n \in \mathbb{N}$$

Proof. Recall that $\tilde{\pi}_t : \mathbb{R}^I \to \mathbb{R}$, $w \mapsto w(t)$.

By definition the sets

$$\{\tilde{\pi}_{t_1} \in A_1, \dots, \tilde{\pi}_{t_n} \in A_n\} \quad t_i \in I, A_i \in \mathcal{B}(\mathbb{R}), n \in \mathbb{N},$$

are a generator of $\mathcal{B}^I(\mathbb{R})$. Therefore

$$C_0(I) \cap \{\tilde{\pi}_{t_1} \in A_1, \dots, \tilde{\pi}_{t_n} \in A_n\} \qquad t_j \in I$$

$$= \{w \in C_0(I) : \tilde{\pi}_{t_1} \in A_1, \dots \tilde{\pi}_{t_n} \in A_n\} \qquad A_j \in \mathcal{B}(\mathbb{R}), \; n \in \mathbb{N}$$

generates $C_0 \cap \mathcal{B}^I(\mathbb{R}) \underset{\substack{\uparrow \\ \text{by previous result}}}{=} \mathcal{B}(C_{C_0}(I))$

The generator above is a π-system and therefore if two probab. measures coincide on it, they coincide on $\left(C_{[0]}(I), \mathcal{B}(C_{[0]}(I)) \right)$. $\lrcorner$

Remark. The σ-Algebra $\mathcal{B}^I(\mathbb{R})$ is "small".
For example it is possible to prove that

- For every $\Gamma \in \mathcal{B}^I(\mathbb{R})$, there exists an at most countable set $S \subset I$ s.t.

  $\hookrightarrow$ ogni insieme in $\mathcal{B}^I$ contiene funzioni che coincidono solo su un insieme numerabile

$$v \in \Gamma,\ w \in \mathbb{R}^I,\ v|_S = w|_S \Longrightarrow w \in \Gamma. \quad (*)$$

it means that the sets in $\mathcal{B}^I(\mathbb{R})$ are big!

- $C_{[0]}(I) \notin \mathcal{B}^I(\mathbb{R})$

  $\hookrightarrow$ perché se v continua allora $\exists w : w|_S = v|_S$ e w non continua perché S numerabile

Sketch of proof. Show that

$$\Sigma' := \left\{ \Gamma \in \mathcal{B}^I(\mathbb{R}) : \Gamma \text{ satisfies } (*) \text{ for an at most countable set } S \subset I \right\}$$

is a σ-Algebra. Check that the family of cylinder sets is in Σ' and conclude that

$$\sigma(\mathcal{G}) (= \mathcal{B}^I(\mathbb{R})) \subset \Sigma'$$

Use the first assertion to prove the second. $\lrcorner$

Now we go back to the construction of a probability space so that the canonical process is a BM. For this, we resort now to the Kolmogorov's Theorem.

Theorem (Kolmogorov). For each $t_1, \dots, t_n \geq 0$, $n \in \mathbb{N}$, with $t_i \neq t_j$ for $i \neq j$ let $p_{t_1, \dots, t_n}$ be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Assume that the family of probability measures is consistent, i.e.,

1) $p_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = p_{t_{\sigma(1)}, \dots, t_{\sigma(n)}}(A_{\sigma(1)} \times \dots \times A_{\sigma(n)})$ for every permutation $\sigma : \{1, \dots, n\} \mapsto \{1, \dots, n\}$ and $A_i \in \mathcal{B}(\mathbb{R})$

2) $p_{t_1, \dots, t_n}(A_1 \times \dots \times A_{n-1} \times \mathbb{R}) = p_{t_1, \dots, t_{n-1}}(A_1 \times \dots \times A_{n-1})$ for all $A_i \in \mathcal{B}(\mathbb{R})$.

Then there exists a unique probability measure μ on $(\mathbb{R}^I, \mathcal{B}^I(\mathbb{R}))$ such that the canonical process $X_t(\omega) = \omega(t)$ has finite dimensional distribution $p_{t_1, \dots, t_n}$ i.e.

$$P(X_{t_1} \in A_1 \dots X_{t_n} \in A_n) = p_{t_1 \dots t_n}(A_1 \times \dots \times A_n)$$

for all $t_i \geq 0$, $A_i \in \mathcal{B}(\mathbb{R})$.

Proof. See References

Corollary. There exists a unique measure μ on $(\mathbb{R}^I, \mathcal{B}^I(\mathbb{R}))$ such that the canonical process $(X_t)_{t \geq 0}$ satisfies: $X_0 = 0$, stationary increments, indep. increments.

Proof. For $t_1 \dots t_n \geq 0$, with $t_i \neq t_j$, $i \neq j$, let $p_{t_1 \dots t_n}$ be the Gaussian distribution with mean vector $\underline{m} = \underline{0}$ and covariance matrix C: $(C)_{ij} = \min(t_i, t_j)$, $i, j = 1, \dots, n$. This probability measure is consistent. Therefore there exists a unique probability measure μ on $(\mathbb{R}^{I}, \mathcal{B}^{I}(\mathbb{R}))$ such that the canonical process $(X_t)_{t\geq 0}$, satisfies

$$(X_{t_1} \cdots X_{t_m}) \overset{d}{=} P_{t_1 \cdots t_m}(\cdot) = N(\underline{0}, C)$$

for all $t_i \geq 0$. The result follows by the following result

Lemma. Let $(X_t)_{t\geq 0}$ be a 1-dim stochastic process. The following are equivalent

(i) $X_0 = 0$, it has stationary and indep increments

(ii) The distribution of the vector $(X_{t_1} \cdots X_{t_m})$, is, for any choice of $t_1 \cdots t_m \geq 0$, Gaussian with mean $\underline{0}$ and covariance matrix C.

In particular

$$E e^{i \langle \xi, \underline{X} \rangle} = e^{-\frac{1}{2}\langle \xi, C\xi \rangle} \qquad \xi \in \mathbb{R}^m.$$

What is missing? Continuity of sample paths!

We resort to the following theorem (Kolmogorov - Čentsov)

Let $(X_t)_{t\geq 0}$ be a d-dimensional stochastic process on a probability space $(\Omega, \mathcal{A}, P)$. If there are constants $c>0$ and, $\alpha, \beta > 0$ s.t.

$$E(|X_t - X_s|^{\alpha}) \leq c\,|t-s|^{1+\beta}, \quad s, t \geq 0,$$

then X_t has a version Y_t with continuous sample paths.

In particular the version can be chosen so that the paths are almost surely locally Hölder continuous of order γ, for $\gamma \in (0, \beta/\alpha)$, i.e., for a.e. $\omega \in \Omega$ there is, for every $T>0$ and $\gamma \in (0, \beta/\alpha)$ some constant $c = c(\gamma, T)$ such that

$$|Y_t^{(\omega)} - Y_s^{(\omega)}| \leq c|t-s|^{\gamma}, \quad s,t \in [0,T]$$

Proof. See references

Corollary. Brownian motion exists.

Proof. We know that the canonical process on $(\mathbb{R}^I, \mathcal{B}^I(\mathbb{R}), \mu)$ satisfies $X_0 = 0$ and has stationary and indep increments. Then, note that

$$\underbrace{E|X_t - X_s|^4 = E|X_{t-s}|^4}_{\text{stationary}} \underbrace{=}_{\substack{\text{self-similarity} \\ X_{t-s} \sim \sqrt{t-s}\, X_1}} |t-s|^2 \underbrace{E|X_1|^4}_{=3} < \infty$$

for all $s,t \geq 0$. Hence the version (say $B_t, t \geq 0$) given by Kolmogorov-Centsov provide a process with continuous paths.

Remark. Let $B = (B_t, t \geq 0)$ be the continuous version. Then

$$(\mathbb{R}^I, \mathcal{B}^I(\mathbb{R}), \mu) \xrightarrow{B} (C_{(0)}(I), \mathcal{B}(C_{(0)}(I)), P)$$

where $P(B) = \mu(\bar{A})$ for any $\bar{A} \in \mathcal{B}^I : \bar{A} \cap C_0 = B$. The canonical process on $(C_0(I), \mathcal{B}(C_0(I)), P)$ is the BM we were looking for.

The d-dimensional BM

It is possible to construct a process fitting the definition of BM in $\mathbb{R}^d$. This is called the d-dimensional BM. Essentially, the procedure is based on the following implications

$B_t := \left(B_t^{(1)} \dots B_t^{(d)}\right)$ is a d-dimensional Brownian Motion

$$\Updownarrow$$

The coordinate processes $\left(B_t^{(j)}\right)_{t \geq 0},\ j = 1 \dots d$ are indep. 1-dim. BMs.

The implications above can be proved, e.g., using characteristic functions.